

Is Application Knowledge Enough to Buy the Right Programmable Power Supply



Sneha Ambastha is a technical journalist at EFY

Being a test engineer, what is the first question you ask yourself when you decide to buy a programmable power supply (PPS)? Is it the price or features you look for? Now, when you think of features, is it only electrical properties like current, voltage or frequency that you think are important? So many questions to be answered, but do you have the answers?

We all know that a power supply that has an internal microcomputer with an option to customise its working as per the tester's requirement is called PPS. Now, the question is, what can be customised? Electrical parameters like voltage, current, frequency, time interval in which output is required, input and output formats can all be customised through different computer-programming languages like C, standard commands for programmable instruments (SCPI), etc.

A PPS has a number of things associated with it, starting from remote-controlled interfaces to different compliance standards. All this makes it very difficult

for the buyer to purchase the correct PPS, without knowing the options in detail.

PPS with different interfaces

A PPS needs to be connected to a computer so that it can be programmed, and for that, it has plenty of interfacing options. With so many available options for interfaces, one can easily decide the right PPS required for testing. Every interface has its advantages that make the selection easy. On the contrary, you will find some PPSs that are compatible only with specific interfaces, like the XPD series from Ametek.

Remote interfaces required

1. RS232
2. GPIB
3. LXI Ethernet (LAN)
4. USB
5. Analogue

RS232. The recommended standard number 232 (RS232) is a 25-pin D-type connector. However, low transmission speed, large voltage swing and large standard connectors affect it.

GPIB. The general-purpose interface bus (GPIB) was created as Hewlett-Packard interface bus (HP-IB) with IEEE 488 standard as a standard digital interface for programmable instrumentation.

LXI Ethernet (LAN). LAN extensions for instrumentation (LXI) is a standard that ensures inter-operability and communicates with systems using the Ethernet. It complements existing test and measurement systems like GPIB.

Isolated analogue. An isolated analogue interface allows remote programming using analogue controls. It not only controls the power supply but also reads back output monitor signals.



Keysights 3631A 80W triple-output power supply